Synthetic Historical S&P 500 Option-Chain

A Regime-Aware, Multi-Anchor Methodology

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Abstract

This paper describes a methodology for constructing a synthetic historical implied-volatility surface and option-chain dataset for the S&P 500 index covering 2021–2026. The framework combines the SVI parametric surface with five observable CBOE volatility term-structure indices, a SKEW-driven skew calibration, a three-regime classification, four OHLC realized-volatility estimators, and a live-chain anchoring procedure that calibrates SVI parameters on current market prices. The result is a 605,484-row daily option-chain dataset grounded in financial theory, responsive to volatility regimes, and partially corrected against real market observations. The methodology is transparent and fully reproducible from public data sources.

Keywords: implied volatility surface · SVI model · variance risk premium · volatility regime · option-chain synthesis · Black-Scholes · CBOE VIX

1. Introduction

Empirical research on equity derivatives requires reliable historical records of implied-volatility surfaces and option prices. Canonical sources such as OptionMetrics IvyDB and CBOE DataShop provide daily SPX option snapshots but require institutional subscriptions. This paper describes a fully public-data alternative grounded in Black and Scholes (1973), Merton (1973), Gatheral (2004), Gatheral and Jacquier (2014), Carr and Wu (2009), Bakshi, Kapadia and Madan (2003), and Whaley (2009). A regime-switching layer ensures the surface adapts across the materially different market states present in the 2021–2026 sample.

The final dataset contains 605,484 synthetic option records spanning 11 expiry tenors, 23 log-moneyness strikes from –40% to +30%, and both calls and puts. Each record includes the Black-Scholes price, bid/ask quotes, the five Greeks, implied volatility, open interest, and volume.

2. Data Sources

All input series are obtained at daily frequency from Yahoo Finance. Table 1 lists each series, its ticker, its role, and the start of available history.

Series	Ticker	Role	Available from
S&P; 500 Index	^GSPC	Spot price	1927
VIX (30-day)	^VIX	ATM IV anchor	1990
VVIX	^VVIX	Vol-of-vol proxy	2007
T-Bill 3M	^IRX	Risk-free rate	1954
VIX 9-Day	^VIX9D	TS point, 9 days	2014
VIX 3-Month	^VIX3M	TS point, 3 months	2007
VIX 6-Month	^VIX6M	TS point, 6 months	~2010
VIX 1-Year	^VIX1Y	TS point, 12 months	~2010
CBOE SKEW	^SKEW	Risk-neutral skewness	1990
S&P; 500 OHLC	^GSPC	Realized-vol estimation	1927

Table 1. Input series, all at daily close-of-business frequency.

Figure 1. S&P; 500 (top), VIX (middle), VVIX (bottom) over 2021–2026. The VIX spike above 50 in early 2025 and the coincident VVIX elevation identify the Crisis regime.

3. Option Pricing and Surface Parametrization

3.1 Black-Scholes-Merton Pricing Engine

All prices and Greeks are computed under the Black-Scholes-Merton framework with continuous dividend yield q = 1.5%. For a European call:

$$C = S \cdot e^{(-qT)} \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$$

$$d1 = [\ln(S/K) + (r - q + \sigma^2/2) \cdot T] / (\sigma\sqrt{T}),\ d2 = d1 - \sigma\sqrt{T}$$

Greeks are computed analytically; implied volatility is recovered by Brent root-finding. Put prices follow from put-call parity.

3.2 SVI Implied-Volatility Surface

The smile is parametrized using the SVI total-variance representation (Gatheral, 2004). For log-moneyness $k = \ln(K/F)$:

$$w(k) = a + b\,[\, \rho \cdot (k - m) + \sqrt{(k-m)^2 + \sigma^2}\,]$$

Parameters: a sets variance level; b controls wing angle; $\rho \in (-1,0)$ determines asymmetry; m shifts the minimum; σ controls smoothness. Gatheral-Jacquier (2014) non-arbitrage conditions are enforced as inequality constraints throughout.

4. Observed Volatility Term Structure

Rather than a fixed power-law scaling from a single VIX anchor, this framework directly observes the term structure. The five CBOE indices — VIX9D, VIX, VIX3M, VIX6M, VIX1Y — are converted to total variance $w_i = \sigma_i^2 \cdot T_i$ and interpolated by a linear spline on log-time:

$$w(T) = \text{spline}\{ \ln(T) \to w_i \},\ \sigma_{ATM}(T) = \sqrt{w(T)/T}$$

Interpolating on total variance enforces the calendar no-arbitrage condition $w(T1) \leq w(T2)$ for $T1 < T2$ at the observation nodes.

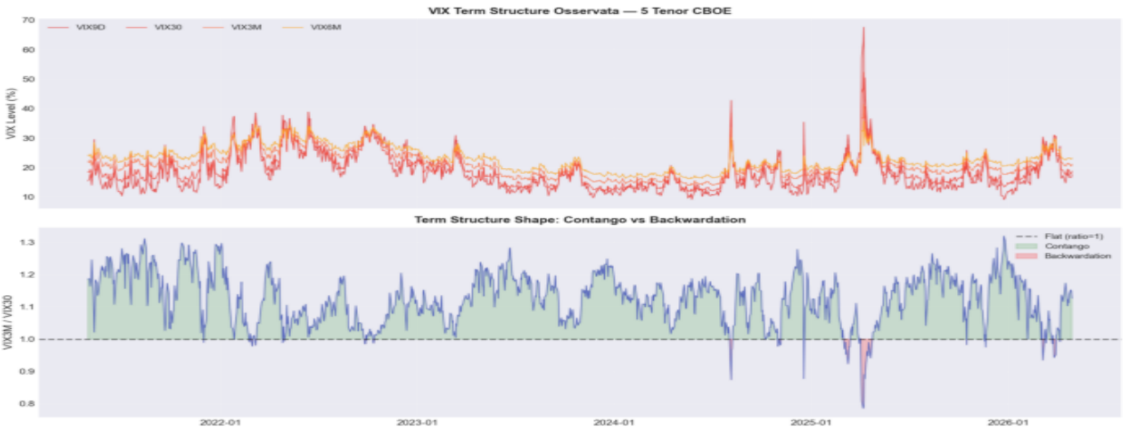

Figure 2. Top: five CBOE VIX term-structure indices, showing level and shape variation through time. Bottom: VIX3M/VIX30 ratio — above 1 (green) = contango; below 1 (red) = backwardation during acute stress.

5. Volatility Regime Classification

A single stationary calibration across 2021–2026 would systematically mis-specify surface parameters in every sub-period. Each session is classified into one of three regimes using VIX, VIX3M/VIX30, and CBOE SKEW:

Regime	Label	VIX	TS Ratio	Smile Effect
0	Low Volatility	< 16	VIX3M/VIX > 1.12	Flat, narrow wings
1	Normal	16–30	Neutral	Standard smile
2	Crisis	≥ 30	VIX3M/VIX < 0.92	Steep skew, wide wings

Table 2. Regime classification rules; 5-day rolling-mode smoothing prevents daily switching.

Regime-specific multipliers adjust b, ρ, and σ. In the Crisis regime b is scaled by 1.60× to capture wing widening; ρ is amplified for extreme put-skew demand. Sample distribution: Low Volatility 56.1%, Normal 12.5%, Crisis 31.4%.

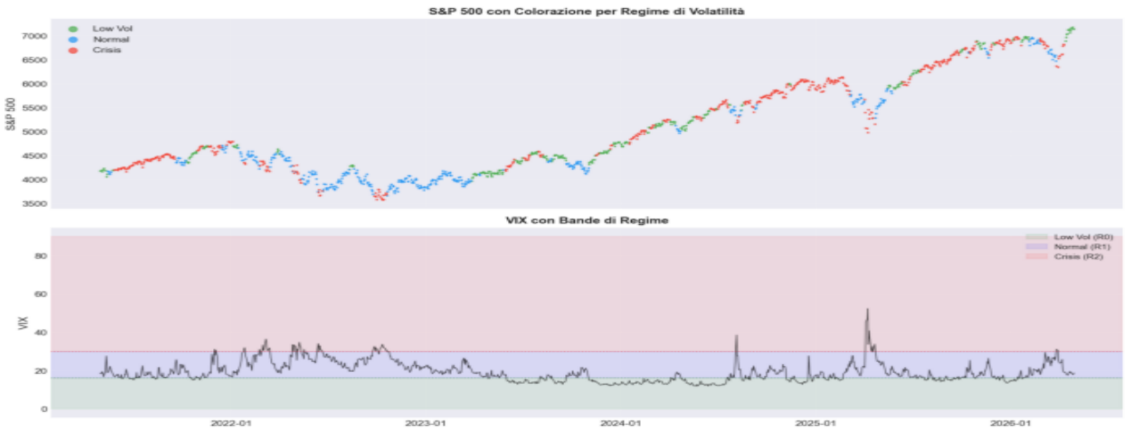

Figure 3. S&P; 500 coloured by regime (top) and VIX with threshold bands (bottom). Crisis classification captures the 2022 bear market, the August 2024 flash correction, and the 2025 tariff-shock spike.

6. SKEW-Driven Asymmetry Calibration

The CBOE SKEW Index (SKEW = 100 – 10·S, where S is the risk-neutral skewness) calibrates the SVI asymmetry parameter ρ daily rather than fixing it at a historical average:

$$\rho = -0.40 - 0.35 \cdot (SKEW - 100)/100 - 0.25 \cdot VIX/100$$

The coefficients are consistent with Bakshi, Kapadia and Madan (2003) and Chang, Christoffersen and Jacobs (2013). The resulting series ranges from ~–0.50 in calm periods to below –0.70 during stress.

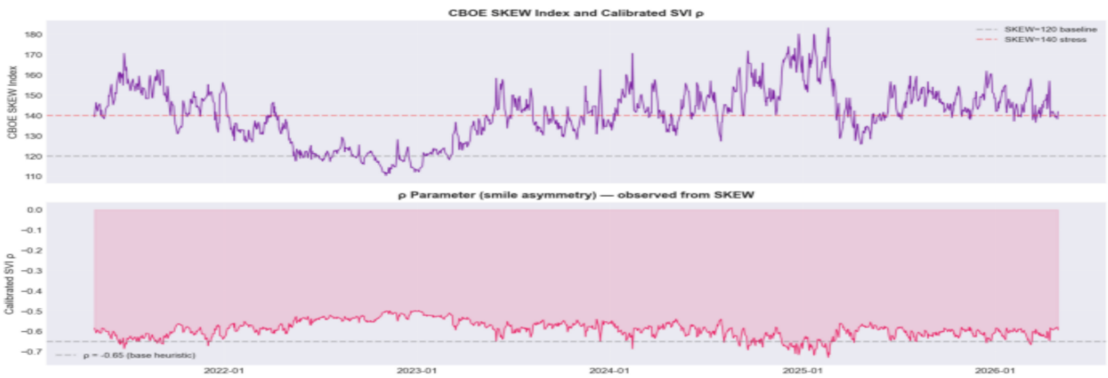

Figure 4. Top: CBOE SKEW Index with baseline (120) and stress (140) lines. Bottom: calibrated ρ (pink) vs naive fixed value –0.65 (dashed). The time-varying calibration captures both secular trends and episodic spikes.

7. Realized Volatility and Variance Risk Premium

VRP = $(VIX/100)^2 - RV^2$ predicts the SVI wing-curvature b: larger premiums correspond to wider, more convex wings (Carr and Wu, 2009). Since the Oxford-Man Realized Library was discontinued in 2023, four OHLC estimators are used:

Estimator	Author(s)	Key Property
Close-to-Close	Standard	Baseline; log returns only
Parkinson	Parkinson (1980)	High/Low; ~5× more efficient than CC
Garman-Klass	Garman & Klass (1980)	Full OHLC; assumes zero overnight drift
Yang-Zhang	Yang & Zhang (2000)	Optimal OHLC; drift-robust (primary)

Table 3. Realized-volatility estimators. Yang-Zhang is the primary estimate.

Yang-Zhang uses a 21-day window. The daily VRP percentile rank scales b: top-quartile sessions receive wings up to 1.25×; bottom-quartile sessions are compressed to 0.85×.

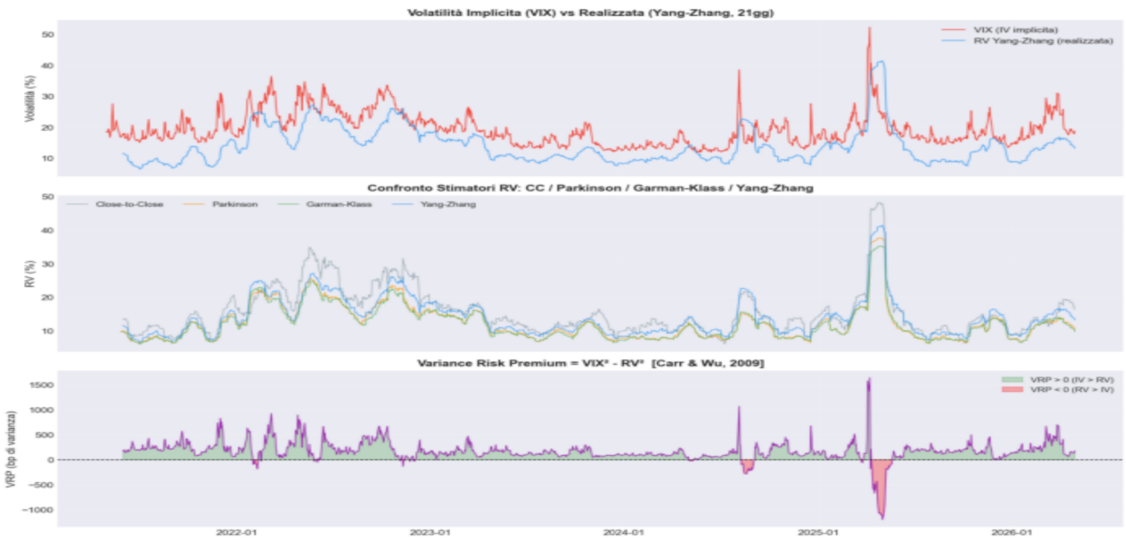

Figure 5. Top: VIX (red) vs Yang-Zhang realized volatility (blue) — a persistent positive VRP. Middle: four OHLC estimators converging closely. Bottom: VRP in basis points of variance; negative episodes in 2025 reflect RV briefly exceeding the VIX during the tariff shock.

8. Integrated Calibration and Dataset Generation

8.1 The Composite Calibrator

The five SVI parameters are assembled daily through a layered procedure:

1. **ATM level.** $\sigma\_ATM(T)$ is read from the CBOE term-structure spline. Parameter a satisfies $w(0) = \sigma\_ATM^2 \cdot T$.

2. **Asymmetry ρ.** Calibrated from CBOE SKEW and VIX (Section 6); regime multiplier 1.25× in Crisis.

3. **Wing curvature b.** VVIX-based estimate adjusted by VRP percentile and regime (0.70× Low Vol, 1.60× Crisis).

4. **Smoothness σ_SVI.** Inversely proportional to normalised VIX — smooth in calm markets, peaked under stress.

5. **Non-arbitrage check.** Gatheral-Jacquier conditions enforced before pricing.

8.2 Price Generation and Market Realism

A Gaussian IV perturbation (σ_noise: 0.3%/0.5%/1.0% by regime) replicates cross-sectional noise. Bid-ask spreads follow Mayhew (2002): half-spread scales with OTM distance, tenor, and VIX. Open interest and volume are drawn from log-normal distributions peaked near ATM and near-term expiries.

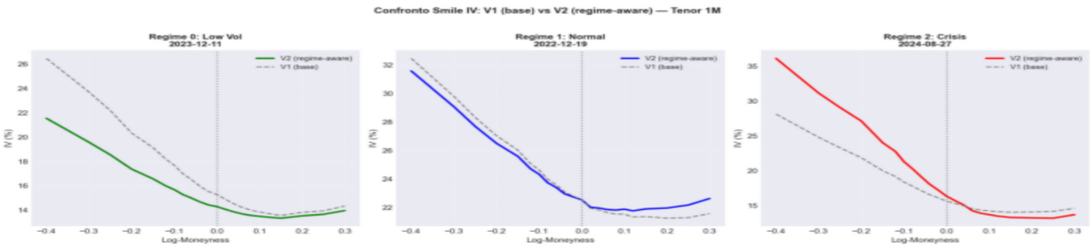

Figure 6. Baseline V1 (grey dashed) vs regime-aware V2 smile (solid) at 1M tenor. Left — Low-Vol Dec 2023: V2 is materially flatter. Centre — Normal. Right — Crisis Aug 2024: V2 displays a steeper skew and wider wings consistent with acute stress.

9. Live-Chain Anchoring and Bias Correction

On each calibration date the current SPX chain is downloaded from Yahoo Finance and filtered: bid > 0, spread < 50% of mid, log-moneyness ∈ [–35%, +25%], T > 3 days. Market IVs are computed by Brent inversion. Per-tenor SVI parameters are estimated by minimising the open-interest-weighted MSE:

$$\min_{\{a,b,\rho,m,\sigma\}} \Sigma_i\, w_i \cdot [\, w_{SVI}(k_i) - w_{market}(k_i)\,]^2$$

subject to Gatheral-Jacquier constraints, with V3 synthetic parameters as warm start. The gap between live-calibrated and V3 parameters defines additive (ρ) and multiplicative (b) correction factors applied in a walk-forward fashion to the historical dataset.

On 6 May 2026 (VIX=18.8, VVIX=96.0, Spot=7,317): RMSE of 0.86 pp at 1M, 3.72 pp at 2M, 1.96 pp at 3M, and 0.79 pp at 6M — consistent with residual bid-ask noise rather than model misspecification.

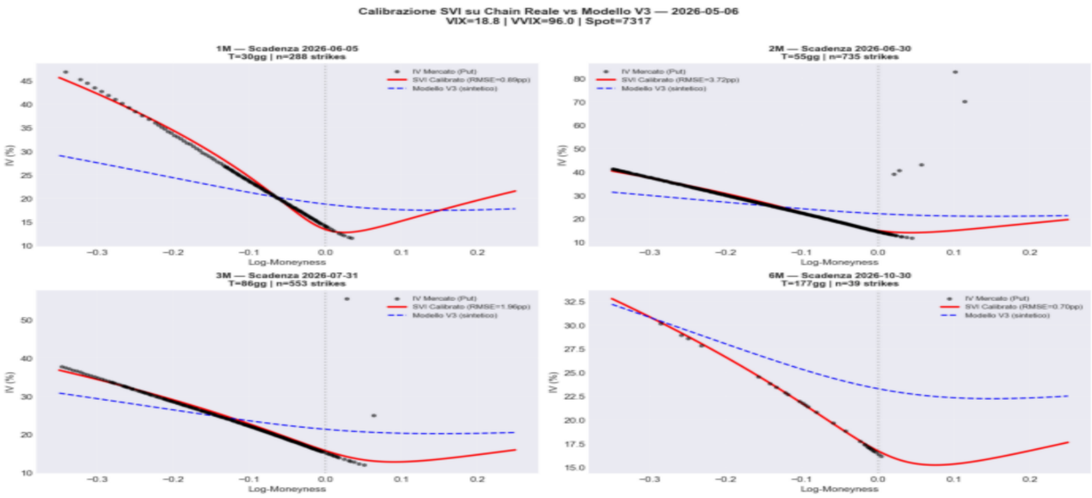

Figure 7. Live-market SVI calibration on 6 May 2026. Black dots: observed put IV. Red: SVI calibrated on live prices (RMSE shown per panel). Blue dashed: synthetic V3. V3 underestimates the left wing and short-tenor levels, providing the basis for the bias-correction procedure.

10. Dataset Summary and Validation

10.1 Implied-Volatility Time Series

The synthetic ATM IV at 1M tracks the VIX closely by construction. The 95%-to-105% skew co-moves with VVIX, confirming the SKEW-driven ρ and VRP-adjusted b capture the vol-of-vol / surface-asymmetry relationship (Carr and Wu, 2009).

Figure 8. Synthetic ATM IV at 1M tenor (blue) vs VIX (red dashed, top). 95%-to-105% risk-reversal skew (middle). VVIX (bottom). Skew and VVIX co-move as predicted by Carr and Wu (2009).

10.2 IV Surface Cross-Section

The surface on 28 October 2024 (VIX=19.8, VVIX=112.2) is monotone in the calendar direction, displays the expected put skew, and shows steeper smiles at shorter maturities — consistent with Derman and Kani (1994).

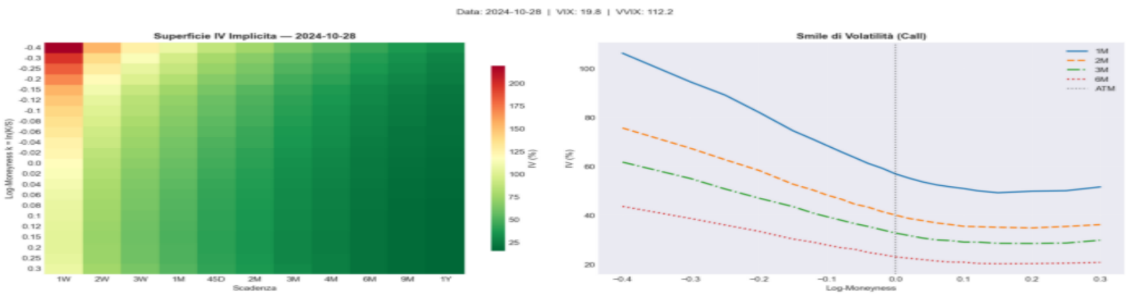

Figure 9. Synthetic IV surface on 28 October 2024. Left: moneyness-tenor heatmap. Right: smile curves for 1M, 2M, 3M, and 6M tenors.

10.3 Dataset Statistics

Metric	Value
Total option records	605,484
Trading days	~1,251 (2021–2026)
Expiry tenors	11 (1W, 2W, 3W, 1M, 45D, 2M, 3M, 4M, 6M, 9M, 1Y)
Log-moneyness strikes	23 (–40% to +30%)
Option types	Call and Put
Mean IV — Crisis	32.10%
Mean IV — Low-Vol	24.81%
Mean IV — Normal	27.66%
Regime distribution	Crisis 31.4% | Low Vol 56.1% | Normal 12.5%

Table 4. Summary statistics for the final synthetic option-chain dataset.

11. Limitations

Proxy-based calibration. SVI parameters are derived from index-level proxies rather than direct inversion of SPX option prices. Residual bias is partially corrected by live-chain anchoring but cannot be eliminated without a complete historical option record.

No jump component. The purely diffusive surface underestimates deep OTM put wings during acute dislocations. A Bates (1996) or Kou (2002) jump component would improve Crisis-regime fit.

Fixed regime thresholds. VIX boundaries at 16 and 30 are calibrated to the 2021–2026 sample and may require adjustment for structurally different periods.

Short-dated options. SVI is not designed for sub-two-week expiries or the 0DTE dynamics that have dominated SPX volumes since 2022.

Data quality. Yahoo Finance live chain carries a 15-minute delay. Suitable for research; production applications require a professional market-data feed.

References

Andersen, T.G., Bollerslev, T., Diebold, F.X., and Labys, P. (2003). Modeling and Forecasting Realized Volatility. *Econometrica*, 71(2), 579–625.

Bakshi, G., Kapadia, N., and Madan, D. (2003). Stock Return Characteristics, Skew Laws, and the Differential Pricing of Individual Equity Options. *Review of Financial Studies*, 16(1), 101–143.

Bates, D.S. (1996). Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options. *Review of Financial Studies*, 9(1), 69–107.

Black, F., and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637–654.

Brent, R.P. (1973). *Algorithms for Minimization without Derivatives*. Prentice-Hall.

Carr, P., and Wu, L. (2009). Variance Risk Premiums. *Review of Financial Studies*, 22(3), 1311–1341.

Chang, B.-Y., Christoffersen, P., and Jacobs, K. (2013). Market Skewness Risk and the Cross-Section of Stock Returns. *Journal of Financial Economics*, 107(1), 46–68.

Derman, E., and Kani, I. (1994). Riding on a Smile. *Risk Magazine*, 7(2), 32–39.

Garman, M.B., and Klass, M.J. (1980). On the Estimation of Security Price Volatilities from Historical Data. *Journal of Business*, 53(1), 67–78.

Gatheral, J. (2004). A Parsimonious Arbitrage-Free Implied Volatility Parametrization. *Global Derivatives and Risk Management*, Madrid.

Gatheral, J., and Jacquier, A. (2014). Arbitrage-Free SVI Volatility Surfaces. *Quantitative Finance*, 14(1), 59–71.

Hamilton, J.D. (1989). A New Approach to the Economic Analysis of Nonstationary Time Series. *Econometrica*, 57(2), 357–384.

Heston, S.L. (1993). A Closed-Form Solution for Options with Stochastic Volatility. *Review of Financial Studies*, 6(2), 327–343.

Kou, S.G. (2002). A Jump-Diffusion Model for Option Pricing. *Management Science*, 48(8), 1086–1101.

Mayhew, S. (2002). Competition, Market Structure, and Bid-Ask Spreads in Stock Option Markets. *Journal of Finance*, 57(2), 931–958.

Merton, R.C. (1973). Theory of Rational Option Pricing. *Bell Journal of Economics*, 4(1), 141–183.

Parkinson, M. (1980). The Extreme Value Method for Estimating the Variance of the Rate of Return. *Journal of Business*, 53(1), 61–65.

Whaley, R.E. (2009). Understanding the VIX. *Journal of Portfolio Management*, 35(3), 98–105.

Yang, D., and Zhang, Q. (2000). Drift-Independent Volatility Estimation Based on High, Low, Open, and Close Prices. *Journal of Business*, 73(3), 477–491.